



Machine Learning and AI

On Campus Exam	Closed Book
Additional Materials: None	Permitted Materials: Approved Non-Programmable Scientific Calculators Permitted (Green)

Guidelines:

- The **duration** of this paper is 2 hours
 - Word Count: No Word Count
 - Write all answers in the answer booklet provided.
 - Do not open the exam paper until instructed by the Invigilator.
 - Use only black pen or pencil.
 - Mobile and electronic devices are not allowed.
 - Do not communicate with other candidates during the exam.
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Exam Instructions:

Answer ALL Questions

Question 1

- (a) List and briefly describe the three primary types of machine learning, providing an example for each.

(6 marks)

- (b) Derive the Maximum Likelihood Estimation (MLE) for a Poisson regression model in the univariate case. Given n independent observations, each observation y_i follows a Poisson distribution, $p(y_i|\lambda_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}$ with mean $\lambda_i = \exp(wx_i)$, where w is the model parameter to be estimated. Show all steps in the derivation, including:

- (i) Setting up the log-likelihood function.

(4 marks)

- (ii) Differentiating the log-likelihood with respect to w .

(4 marks)

- (iii) Explain why solving the resulting equation for w requires nonlinear optimization.

(4 marks)

- (iv) Briefly explain the role of link functions in Poisson regression and explain why $\log(\lambda_i) = wx_i$ is a suitable choice.

(4 marks)

(Total 22 marks)

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Question 2

- (a) Describe how Bayesian inference incorporates uncertainty in machine learning.

(3 marks)

- (b) For a binary classification problem, explain when you might prefer a Naive Bayes model over a Logistic Regression model.

(4 marks)

- (c) A sensor in a manufacturing plant measures the temperature of a reactor, modelled as a Gaussian random variable T with mean $\mu = 100^\circ\text{C}$ and variance σ^2 . The following formulas and constraints are given to guide your derivation: Differential entropy: $H[T] = -\int_{-\infty}^{\infty} p(t) \ln p(t) dt$, and Gaussian distribution: $p(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right)$, with the constraints: $\int_{-\infty}^{\infty} p(t) dt = 1$ and $\int_{-\infty}^{\infty} (t - \mu)^2 p(t) dt = \sigma^2$

- (i) Derive the entropy $H[T]$ explicitly, and explain how $H[T]$ depends on σ^2 .

(9 marks)

- (ii) Conclude the findings in part (i) by discussing its implications in the context of temperature measurement uncertainty.

(3 marks)

(Total 19 marks)

Question 3

- (a) Consider a Bayesian model with a posterior distribution $p(\theta|x)$ that is intractable to compute directly. To approximate the posterior, we use a variational distribution $q(\theta)$ and the Evidence Lower Bound (ELBO) as the objective function.
- (i) Derive the ELBO starting from the marginal log-likelihood $\log p(x)$, and show how it relates to the KL divergence $\text{KL}(q(\theta)||p(\theta|x))$.
(7 marks)
- (ii) Explain why maximizing the ELBO is equivalent to minimizing the KL divergence between $q(\theta)$ and $p(\theta|x)$.
(2 marks)
- (iii) The ELBO can be expressed as two components: the data term and the regulariser. Write the ELBO in this form and explain the significance of each term in variational inference.
(4 marks)
- (b) Given a target distribution $p(x) \propto e^{-x^2}$ and a proposal distribution $q(x) = \mathcal{N}(0, 1)$, describe how to perform rejection sampling to generate samples from $p(x)$.
(5 marks)

(Total 18 marks)

Question 4

- (a) Define sequential data and explain the difference between stationary and non-stationary sequential data. Provide one example for each type.

(5 marks)

- (b) Consider an HMM modeling the behavior of a robot navigating a building. The robot's **states** represent its location: $S = \{\text{Room1}, \text{Room2}\}$, and its **observations** correspond to actions: $O = \{\text{MoveForward}, \text{TurnLeft}, \text{PickUpItem}\}$. The parameters of the HMM are:

- Transition Matrix:

$$A = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$$

- Emission Matrix:

$$B = \begin{bmatrix} 0.3 & 0.4 & 0.3 \\ 0.2 & 0.6 & 0.2 \end{bmatrix}$$

- Initial Probabilities:

$$\pi = [0.5, 0.5]$$

Use the Forward Algorithm to calculate the likelihood $P(O|\lambda)$ for the observation sequence $O = \{\text{MoveForward}, \text{TurnLeft}, \text{PickUpItem}\}$. Show your calculations (up to four decimal places) step by step.

(14 marks)**(Total 19 marks)**

Question 5

- (a) Explain the concept of a Markov Decision Process (MDP). How does it relate to reinforcement learning?

(6 marks)

- (b) Consider an agent navigating a grid environment with the following states and transitions. The agent can take two actions (a_1 : move up, a_2 : move right). Below are the rewards and state transitions for the given scenario:

Time Step	Current State	Reward	Action	Transition State
1	s_1	+3	a_2	$s_1 \rightarrow s_2$
2	s_2	+5	a_1	$s_2 \rightarrow s_3$
3	s_3	-2	a_2	$s_3 \rightarrow s_4$
4	s_4	+10	a_1	$s_4 \rightarrow s_1$

The Q-learning parameters are:

- Initial Q -values = 0 for all state-action pairs.
 - Learning rate (α) = 0.8.
 - Discount factor (γ) = 0.9.
- (i) Using the Q-learning update rule, compute the Q -values for all state-action pairs after four time steps and draw the updated Q table.

(10 marks)

- (ii) The agent is tasked with identifying the optimal policy for the grid environment based on the Q -values calculated in part (i). The policy is defined as selecting the action with the highest Q -value for each state. Explain how reducing the discount factor from 0.9 to 0.5 would affect the agent's policy and decision-making focus. Provide a conceptual explanation without recalculating the Q -values.

(6 marks)**(Total 22 marks)**